Objectives

- Build a messenger
- Communication is encrypted
- Allow sending messages to offline peers
- Optional: File-Sharing, Group Chats

Technical Challenges

- Connection management
- Message Routing
- Username/ID $\longrightarrow$ IP mapping
- Verifiable peer identification
- Caching messages for offline peers
- Protecting from attackers (Spam, Malicious Peers)
- Reduction of publicly visible metadata (who communicates with whom and when?)